

Yonatan Tesfagiorgis

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EDUCATION

University of Houston, Houston, TX

Expected May 2026

Bachelor of Science, Computer Science

Texas State Technical College, Rosenberg, TX

July 2019

Associate of Science, Cybersecurity

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, PHP, SQL, HTML, CSS, TCP/IP

Backend: Node.js, Express.js, Flask, REST APIs

Frontend: React.js, Vite

Databases: MySQL, FAISS (vector search)

AI/ML: PyTorch, TensorFlow, scikit-learn, NLP, Hugging Face, Transformers, BERT, Docker, Llama, LangGraph, LlamaIndex, LLMs, Pandas, NumPy, Matplotlib, R, Docker, OpenAI, Claude, Git, GitHub, Gitlab, Azure, CI/CD

Concepts: RAG systems, embeddings, prompt engineering, data structures, model evaluation

PROJECT EXPERIENCE

GitHub: <https://github.com/ravenfire24>

University of Houston

NLP Aspect - Based Sentiment Retrieval System (Python, scikit-learn, NLTK, BERT)

August 2025 – November 2025

- Built NLP pipeline for aspect-based sentiment classification using 10,000+ Amazon reviews
- Combined Logistic Regression baseline with BERT embeddings to improve classification performance
- Applied cosine similarity with sentence-transformers for semantic review matching
- Improved precision over keyword-based sentiment classification approaches

AI-Powered Document Assistant (RAG+LLMs) (LangChain, Hugging Face, FAISS, Streamlit)

January 2026 – March 2026

- Built a full-stack Retrieval-Augmented Generation (RAG) system enabling conversational querying over PDF documents
- Implemented semantic search pipeline using sentence-transformer embeddings and FAISS vector database
- Integrated Hugging Face LLMs with prompt engineering to improve response relevance and reduce hallucinations
- Deployed interactive chat interface on Streamlit via Hugging Face Spaces for real-time document exploration

Volunteer Management Application (Full-Stack: Python/Flask, React, MySQL)

June 2025 – August 2025

- Developed a full-stack web application with RESTful APIs for authentication, event scheduling, and user management
- Built responsive React dashboard with real-time analytics for tracking 50+ volunteer events
- Designed MySQL database schema and optimized queries for efficient data retrieval
- Implemented automated report generation (PDF/CSV export) for operational tracking

Chest X-Ray Pneumonia Detection (Python, PyTorch, DenseNet121)

January 2026 – March 2026

- Developed CNN model to classify chest X-rays into normal, pneumonia, and tuberculosis categories using 15,000+ images
- Fine-tuned DenseNet121 with class-weighted loss, achieving 85%+ validation accuracy
- Applied data augmentation techniques to improve generalization and reduce overfitting
- Evaluated performance using confusion matrix and prepared model for deployment

WORK EXPERIENCE

Production Assembly Operator – Randstad

September 2020 – December 2021

- Supervised 5 – 7 production line workers and coordinated workflow execution under time constraints
- Improved operational efficiency by 15% through process optimization and bottleneck reduction
- Trained employees in safety and quality standards, reducing incidents by 25%
- Reduced equipment downtime by 20% through proactive monitoring and maintenance response

LEADERSHIP EXPERIENCE

University of Houston- CougarCS Organization

January 2024 – May 2026

- Led C++ workshops and technical problem-solving sessions for 50+ students
- Mentored peers in programming fundamentals and algorithmic thinking